

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – AI Agent for Smart Farming Advice

Domain of Project – Agriculture

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Recommended font style and size:

Font Style: Arial

Font Size (Heading): 28

Font Size (Content): 20

Problem Statement - 14: AI Agent for Smart Farming Advice

The Challenge

Agriculture is the backbone of food security, yet farmers often struggle with low crop yields, unpredictable weather, pest infestations, soil degradation, and limited access to timely expert advice. Information about crop cultivation, fertilizers, irrigation, government schemes, and market prices is scattered across multiple unreliable sources. Without personalized, real-time guidance, many farmers fail to make informed decisions, resulting in reduced productivity and income.

The Objective

Build an **AI-powered Smart Farming Agent** that delivers **personalized agricultural insights and guidance** for farmers, agricultural professionals, and agribusinesses. The solution should:

Personalized Crop Advisory – Recommend suitable crops, seeds, fertilizers, irrigation schedules, and harvesting timelines based on soil type, weather, location, season, and farming goals.

Agriculture Knowledge RAG Retrieval – Fetch and summarize reliable information on crop cultivation, soil management, pest control, government schemes, and best farming practices from trusted agricultural sources.

.

Multi-Agent System –

- **Crop Advisory Agent** (*personalized crop and cultivation recommendations*)
- **Weather & Irrigation Agent** (*weather forecasting and irrigation planning*)
- **Pest & Disease Detection Agent** (*crop health analysis using text and images*)
- **Market & Government Agent** (*market prices, government schemes, and subsidy information*)
- **Visualization Dashboard** – Show crop health status, weather forecasts, irrigation schedules, soil insights, market trends, and farming progress through an interactive dashboard.

Proposed Solution -

The proposed solution is an AI-powered Smart Farming Agent built using IBM Bob, IBM Granite Models, IBM watsonx.ai, and Retrieval-Augmented Generation (RAG). The platform assists farmers throughout the crop lifecycle by providing intelligent, personalized, and real-time agricultural recommendations.

The system integrates weather forecasting, soil health analysis, pest detection, government agricultural schemes, and market price predictions into a unified AI platform.

The solution consists of four intelligent agents:

Crop Advisory Agent

- Recommends suitable crops
- Suggests seeds
- Advises fertilizers
- Provides irrigation schedules
- Recommends harvesting time

Weather & Irrigation Agent

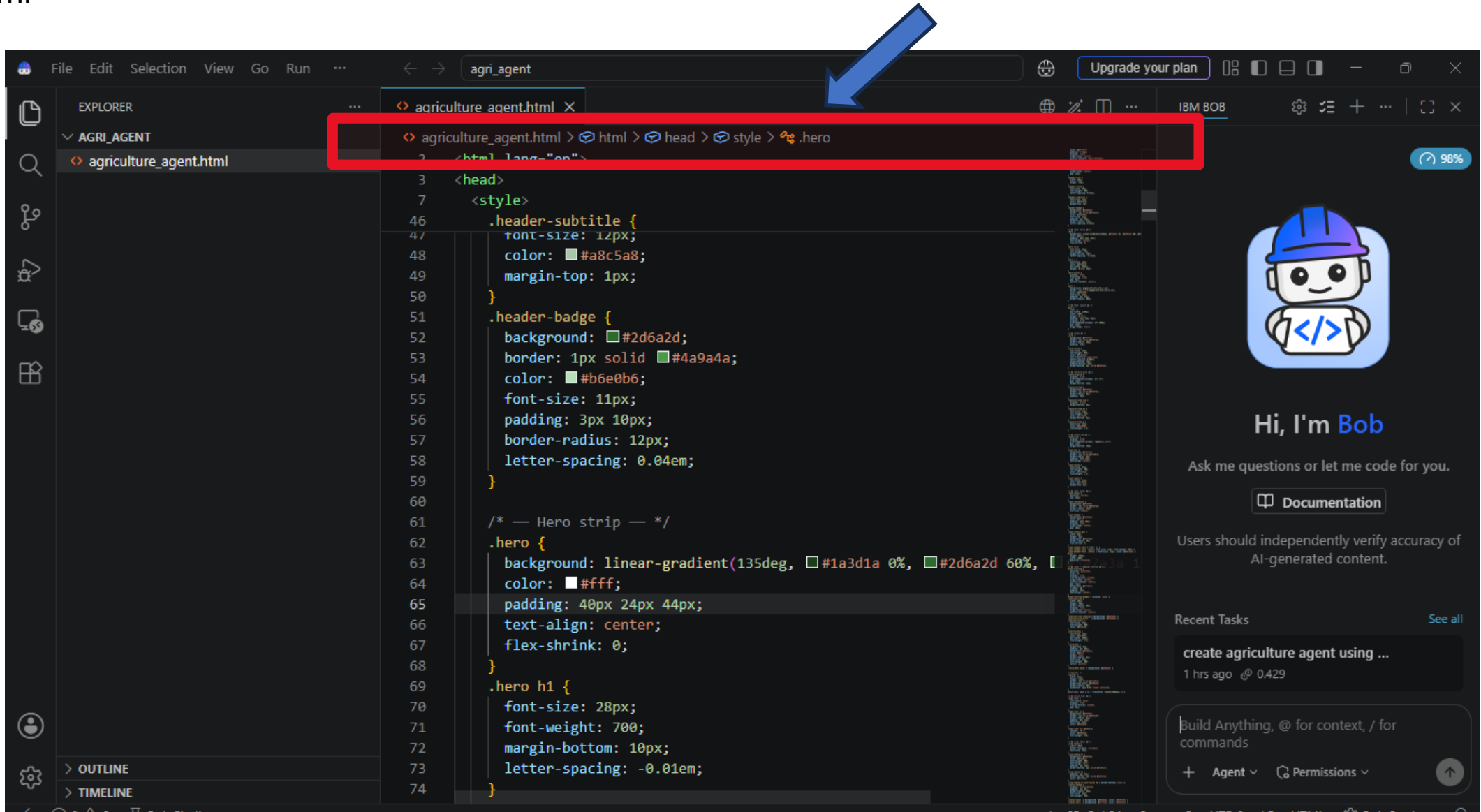
- Retrieves live weather forecasts
- Predicts rainfall
- Generates irrigation schedules
- Issues weather alerts

Technology Used

- **IBM Bob platform** -For designing and orchestrating multi-agent workflows visually
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **OpenWeather API – Weather Forecast**
- **Soil Health APIs / Dataset**
- **Plant Disease Detection Model**
- **Market Price APIs**

Required Files created using IBM bob -

1. Index.html



IBM Bob Screenshots:

The screenshot displays the IBM Bob IDE interface. The main editor window shows a file named `.env` with the following configuration:

```
1 # IBM Watsonx.ai Configuration
2 # Get your API key from: https://cloud.ibm.com/iam/apikeys
3 IBM_CLOUD_API_KEY= L8uheXA-aFuc2l09kBr2ejoJsJP8SIY_1M-1NtFXs__C
4
5 # IBM Watsonx.ai Project Configuration
6 WATSONX_PROJECT_ID= 49af698b-2be0-4ff3-b4fa-c9136e7b03f1
7 WATSONX_URL=https://us-south.ml.cloud.ibm.com
8
9 # Flask Configuration
10 FLASK_SECRET_KEY=your_secret_key_here_change_in_production
11 FLASK_ENV=development
12 FLASK_DEBUG=True
13
14 # Application Settings
15 PORT=5000
16 HOST=0.0.0.0
```

A red circle with the number '2' is positioned over the `HOST=0.0.0.0` line. The sidebar on the right features the IBM Bob logo (a blue robot head with a hard hat) and the text "Hi, I'm Bob". Below this, it states: "I'm your AI development partner. I can augment your workflows to help you understand, plan, and improve real codebases with confidence." A button labeled "Log in to Bob" is visible. At the bottom of the sidebar, there is a "Sign in failed: Cancelled by user." message.

The bottom status bar shows "Screen Reader Optimized" and "Ln 16, Col 13". The Windows taskbar at the very bottom displays the date "26-06-2026" and time "12:14".

File Edit Selection View Go Run Terminal Help

← → bob test

requirements.txt .env.example app.py 8 X index.html script.js README.

Upgrade your plan

EXPLORER

BOB TEST

.venv

static

script.js

style.css

templates

index.html

.env.example

.gitignore

app.py 8

DEPLOYMENT.md

Procfile

README.md

requirements.txt

runtime.txt

SETUP_GUIDE.md

app.py

get_watsonx_model

73 # =====

74 # IBM Watsonx.ai Configuration

75 # =====

76

77 def get_watsonx_model() -> Model | None:

78 """Initialize and return IBM Watsonx.ai model"""

79 try:

80 credentials: Credentials = Credentials(

81 url=os.getenv('IBM_WATSONX_URL', 'https://au-syd.ml.cloud.ibm.com'),

82 api_key=os.getenv('IBM_CLOUD_API_KEY', ' '),

83)

84

85 project_id: str | None = os.getenv('IBM_WATSONX_PROJECT_ID')

86

87 model: Model = Model(

88 model_id='ibm/granite-guardian-3-8b',

89 params=

90 GenParams.DECODING_METHOD: "greedy",

91 GenParams.MAX_NEW_TOKENS: 900,

92 GenParams.MIN_NEW_TOKENS: 50,

93 GenParams.TEMPERATURE: 0.7,

94 GenParams.TOP_K: 50,

95 GenParams.TOP_P: 1

96 },

97 credentials=credentials,

98 project_id=project_id

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

BOB FINDINGS

requests==2.31.0

- urllib3==2.7.0

+ urllib3==2.1.0

+ werkzeug==3.0.1

+ zipp==4.1.0

warning: 'ibm-watson-machine-learning==1.0.335' is yanked (reason: "Causes errors on py 3.9")

(.venv) PS D:\OneDrive - Edunet Foundation\Desktop\bob test>

CODEX

IBM BOB

Create an AI-powered Nutrition ... 42k / 200k 1.17 X

All tasks completed! 13/13

1. Install dependencies: pip install -r requirements.txt

2. Copy .env.example to .env and add IBM credentials

3. Run: python app.py

4. Open: http://localhost:5000

Customization:

The AGENT_INSTRUCTIONS section in app.py allows you to easily customize:

Agent tone and personality

Diet specializations

Indian food preferences

Safety rules and limitations

Response formatting

All files are production-ready with comprehensive error handling, security best practices, and deployment configurations for Heroku, IBM Cloud, AWS, and Docker!

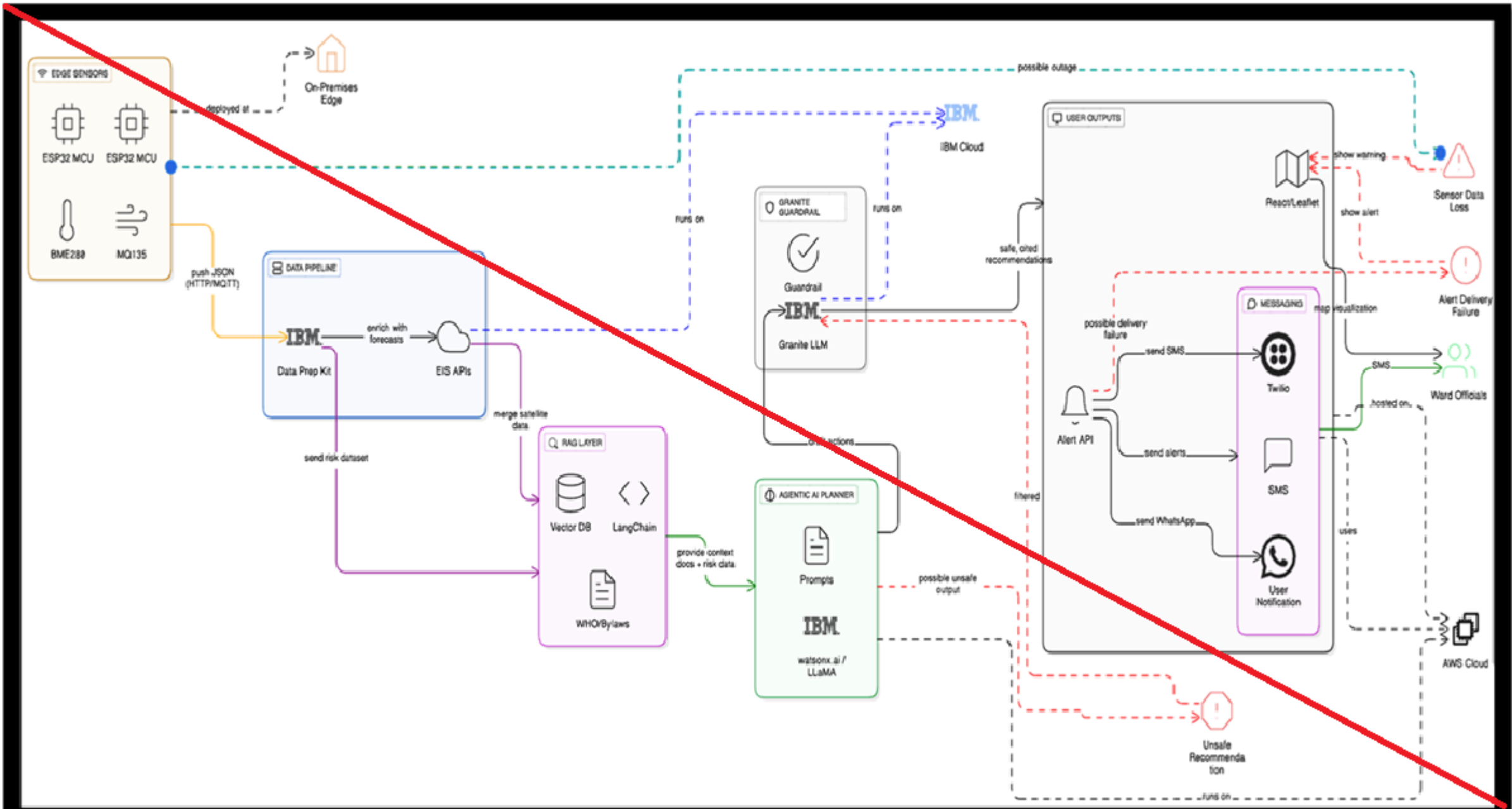
Start New Task

What's on your mind?

(@ to add context, / for commands, hold shift to drag in files/images)

Code

Architecture blueprint for IBM bob:



Role of Agentic AI in the solution

Agentic AI enables the Smart Farming Agent to function as an autonomous agricultural advisor rather than a simple chatbot.

Each specialized agent independently performs dedicated tasks while collaborating with other agents.

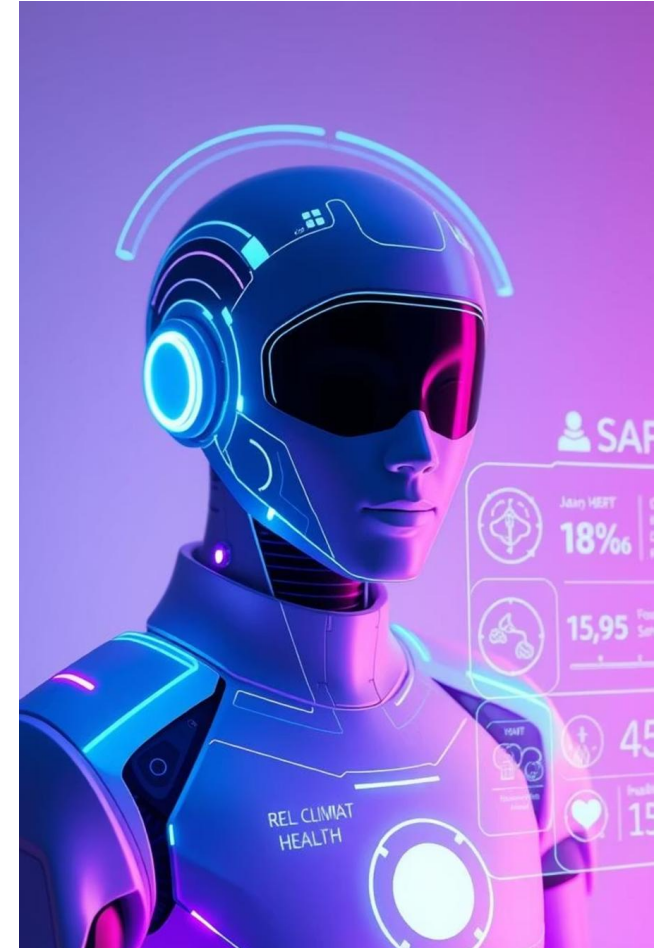
The Crop Advisory Agent recommends suitable crops and farming practices.

The Weather & Irrigation Agent continuously monitors weather conditions and optimizes irrigation.

The Pest Detection Agent analyzes crop images to identify diseases and recommend treatments.

The Market Insights Agent forecasts crop prices and identifies profitable markets.

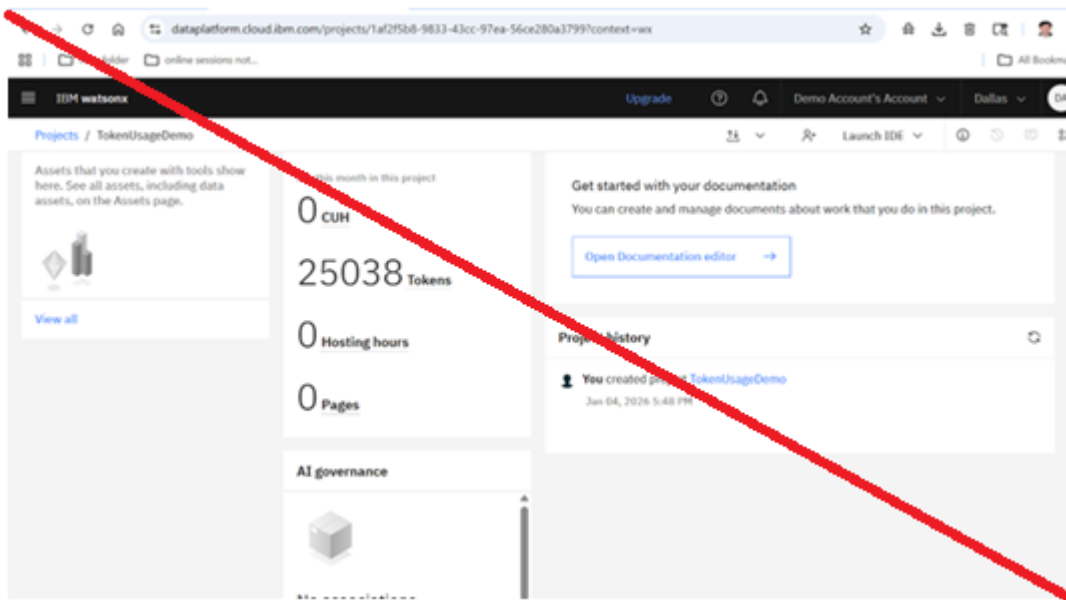
Agentic AI continuously adapts recommendations based on farmer location, weather updates, crop stage, and historical farm data, enabling proactive and personalized farming assistance.



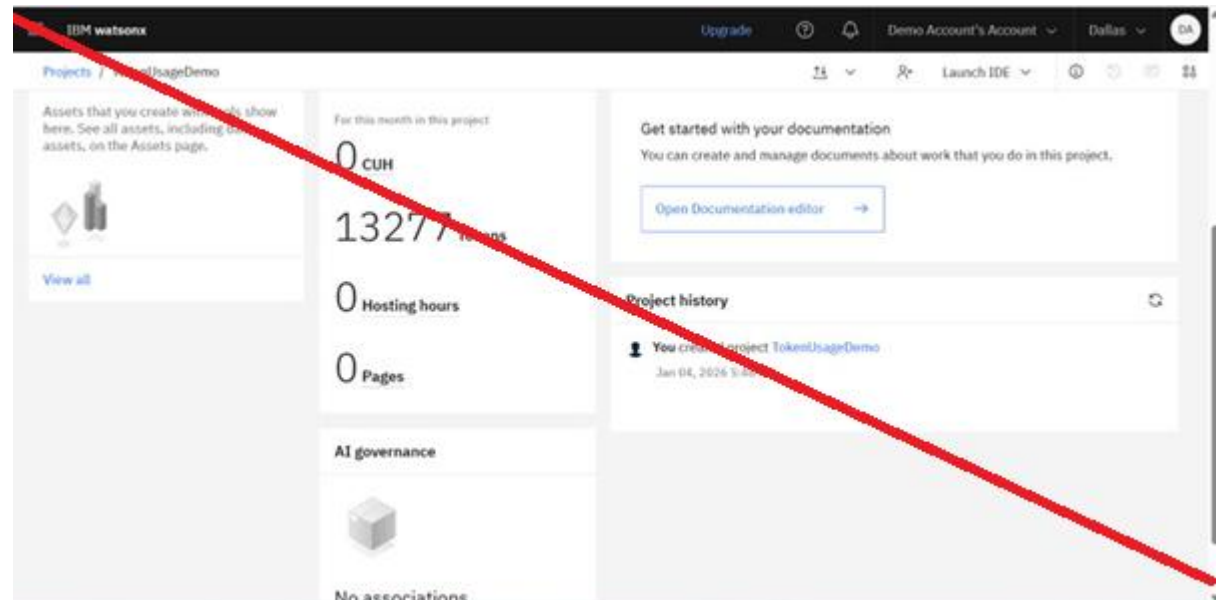
Token Usage

11,761 token are used

Before Prompt



After the Prompt



Project Output Screenshot

The screenshot displays the AgriAgent web application in a browser window. The browser's address bar shows the file path: `C:/Users/Rukmini/Downloads/agri_agent/agriculture_agent.html`. The application header includes the AgriAgent logo, the tagline "AI-Powered Agriculture Assistant", and an "IBM watsonx" badge. The main banner features the heading "Smart Farming, Smarter Decisions" and a subtext: "Ask your AI agriculture assistant about crop health, soil conditions, weather planning, pest control, and sustainable farming practices." Below the banner are six category buttons: "Crop Advisory", "Soil Analysis", "Weather Insights", "Pest & Disease", "Yield Forecasting", and "Irrigation Planning".

The dashboard contains several key metrics and sections:

- 98% Advisory Accuracy**
- 50+ Crop Types**
- 24/7 AI Support**
- AgriAgent AI Assistant** (Online - Ready to help)
- Crop Monitoring**: Real-time insights on crop growth stages, nutrient deficiencies, and harvest timing.
- Soil Intelligence**: pH, moisture, and fertility analysis with personalised fertiliser recommendations.
- SUGGESTED QUESTIONS**:
 - What fertiliser is best for wheat at grain-fill stage?
 - How do I identify early signs of leaf rust?
 - Recommend an irrigation schedule for sandy soil.

The bottom of the image shows a Windows taskbar with the system clock at 13:44 on 22-07-2026, a weather widget showing 22°C and Cloudy, and various application icons.

Project Output Screenshot

**AgriAgent**
AI-Powered Agriculture Assistant

← Back to Home

 New Chat

 Clear Chat

Chat History

Today

How can I improve soil fertility naturally?
10:32 AM

Best irrigation for sandy soil?
10:25 AM

Tomato leaf curl problem
10:18 AM

 **Tip**

Ask specific questions for better and more accurate advice.

Chat with AgriAgent

Get expert advice for your farming queries

● Online

How can I improve soil fertility naturally?
10:32 AM



Improving soil fertility naturally is essential for sustainable farming. Here are some effective methods:

- Organic Matter:** Add compost, farmyard manure (FYM), or green manure to increase soil organic content.
- Crop Rotation:** Rotate crops to prevent nutrient depletion and improve soil structure.
- Green Manuring:** Grow plants like sunhemp or dhaincha and plow them into the soil.
- Biofertilizers:** Use rhizobium, azotobacter, and phosphobacteria to enhance nutrient availability.
- Mulching:** Cover soil with organic mulch to retain moisture and improve soil health.

These practices help maintain long-term soil fertility and improve crop productivity.

10:32 AM

Best organic fertilizer?

How to test soil health?

Benefits of crop rotation?

Green manure examples?

Type your question here...

Send 

Project Output Screenshot

**AgriAgent**
AI-Powered Agriculture Assistant

AgriAgent AI Assistant

Online · Ready to help

← Back to Home

New Chat

Clear Chat

Chat History

Today

Which fertilizer to use for onion?
11:20 AM

Irrigation schedule for onion crop
10:58 AM

Onion thrips control methods
10:35 AM

 **Tip**

Ask specific questions for better and more accurate advice.

Chat with AgriAgent

Get expert advice for your farming queries

Online

Which fertilizer should I use for onion crop?
11:20 AM



For a healthy onion crop with good bulb size and yield, use a balanced fertilization approach:

 **1. Basal Dose (At Sowing/Transplanting)**

- **FYM/Compost:** 20–25 tons per acre
- **NPK: 10:26:26** @ 50 kg per acre
- **Micronutrients:** Zinc Sulphate @ 10 kg/acre

 **2. Top Dressing**

- **Urea (46% N):** 25–30 kg per acre in 2–3 split doses
 - First at 25–30 days after transplanting
 - Second at 45–50 days after transplanting

 **3. Micronutrients (If deficiency observed)**

- Spray 0.5% Zinc Sulphate + 0.25% Boron at 30 and 45 days after transplanting

 **Tips:**

- Avoid excess nitrogen, it may increase leaf growth but reduce bulb size.
- Balanced potash (K) helps in better bulb development and shelf life.

Would you like a complete fertilizer schedule for onions?
11:20 AM

Complete fertilizer schedule

Organic options for onion

Micronutrient deficiency in onion

Best NPK ratio for onion

Type your question here...

Send 

Novelty and Uniqueness

- Multi-Agent AI Architecture
- RAG-Based Agricultural Knowledge
- Real-Time Weather Intelligence
- AI-Based Pest & Disease Detection
- Personalized Crop Recommendations
- Government Scheme Integration
- Soil-Based Fertilizer Recommendation
- Market Price Forecasting
- Sustainable Farming Practices
- IBM Granite Powered Decision Making

Git Hub Link

· Please test and Paste the working Github URL – **https://github.com/kalyani-jadhav-project/Agri_AI_Agent**

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Future Scope

- Drone-based Crop Monitoring
- IoT Sensor Integration
- Satellite Image Analysis
- AI Voice Assistant in Regional Languages
- Blockchain-based Crop Traceability
- Smart Irrigation using IoT
- Farm Equipment Recommendations
-

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